

Illumina Stranded Total RNA Prep with Ribo-Zero Plus

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Version	Enforcement Date
1.0	2022-09-14
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REVISIONS FROM THE ORIGINAL VERSION

Version	Summary of modifications	Date and initials
V1.1	Added more input recommendations	14/08/2023

Date: _____

Project: _____

Library prepared by: _____

Purpose: This protocol uses the Illumina Stranded Total RNA Prep with Ribo-Zero Plus kit to deplete abundant total RNA transcripts from cells or tissue and converts the remaining RNA into dual-indexed libraries for sequencing on Illumina platforms.

Illumina Stranded Total RNA Prep with Ribo-Zero Plus

Prepwork:

- Sort the kits out as outlined below
- Replenish consumables at all benches
- Prepare batch excel sheet and print MM vols
- RNAClean XP – 2mL aliquots
- AMPure XP – 2mL aliquots

Kits

Illumina Ribo-Zero Plus rRNA Depletion Kit (Pre-room -20°C)

- ☐ DB1
- ☐ DP1
- ☐ ELB
- ☐ PRB
- ☐ PRE
- ☐ RDB
- ☐ RDE

Illumina cDNA Synthesis kit (Pre-room -20°C)

- ☐ EPH3
- ☐ FSA
- ☐ RSB
- ☐ RVT
- ☐ SMM

Illumina RNA Prep – Ligation (Pre-room -20°C)

- ☐ ATL4
- ☐ EPM
- ☐ LIGX
- ☐ RSB
- ☐ STL

IDT for Illumina RNA UD Indexes – Ligation (Post-room -20°C)

- ☐ RNA Index Anchors
- ☐ DNA/RNA UD Indexes (UDP0XXX)

Consumables

- ☐ AMPure XP beads
- ☐ RNAClean XP beads
- ☐ NF water
- ☐ 8-well strips
- ☐ P10, 20, 200, 1000 tips
- ☐ Reagent reservoir
- ☐ 1.5mL & 2mL tubes
- ☐ 15 & 50mL falcons
- ☐ 5, 10, 25mL pipettes
- ☐ 96-well PCR plate, non-skirted
- ☐ PCR plate sealing film
- ☐ 100% EtOH (Pre and post-room)
- ☐ QC - D1000 tape, BR DNA Qubit
- ☐ Qubit tubes
- ☐ HeLa-S3 total RNA control (Thermo/AM7852)
- ☐ **Seraseq FFPE control**
- ☐ Kimwipes, gloves, EtOH, RNase Away

Illumina Stranded Total RNA Prep with Ribo-Zero Plus

Input:

High quality RNA samples: 10-100 ng total RNA ($RIN \geq 9$) and ($DV200 > 55\%$) samples in 11 μ L.

1-1000 ng purified total RNA ($RIN \geq 9$).

Recommended input for Fresh/Frozen Tissue- 250ng

For FFPE $RIN > 2$ and a $DV200 > 30\%$ with a minimum concentration of at least 100ng/ μ L have a better library prep success rate.

Recommended: Tumor cellularity of 35% or above is recommended for tissue/FFPE samples before choosing the specimens.

Input 250ng-500ng for FFPE

NOTE: Remove enzymes from -20 immediately before adding to master mix, and immediately put back.

NOTE: Vortex and quick spin reagents (except enzymes) before use.

NOTE: When mixing reactions by pipetting using the multichannel, increase the total volume on the pipette to ensure adequate mixing. Undermixing will result in failure of the library prep.

PRE-PCR ROOM

1. Deplete rRNA

☐ From rRNA depletion kit, thaw:

- DB1 (-20°C, blue cap) – thaw at room temp, vortex to mix
- DP1 (-20°C, blue cap) – thaw at room temp, vortex to mix
- RDB (-20°C, yellow cap) – thaw at room temp, vortex to mix
- RDE (-20°C, yellow cap) – store until needed, flick to mix

1.1 Hybridize Probes

- ☐ Dilute 10-100 ng total RNA in NF H₂O to 11 µL in a PCR plate
- ☐ Preheat thermocycler (HYB_DP1)
- ☐ In 1.5 mL tube on ice, prepare Hybridize Probe Master Mix (see excel template)
 - DB1: 3.6 µL / sample
 - DP1: 1.2 µL / sample
 - Pipette MM to mix
- ☐ Add 4 µL Hybridize Probe MM to each well containing RNA
- ☐ Pipette to mix (15 µL total), seal, quick spin
- ☐ Thermocycler HYB_DP1 program (15 min)
 - Lid preheated to 100°C
 - 95°C for 2 min
 - Decrease 0.1°C per second until 37°C, hold at 37°C
- ☐ Quick spin
- ☐ Preheat thermocycler (RNA_DEP)

1.2 Deplete rRNA

- ☐ In 1.5 mL tube on ice, prepare rRNA Depletion Master Mix (see excel template)
 - RDB: 4.8 µL / sample
 - RDE: 1.2 µL / sample
 - Pipette MM to mix
- ☐ Add 5 µL rRNA Depletion MM to each well
- ☐ Pipette to mix (20 µL total), seal, quick spin
- ☐ Thermocycler RNA_DEP program (17 min)
 - Preheat lid to 100°C
 - 37°C for 15 min
 - Hold at 4°C
 - From the rRNA depletion kit, prepare:
 - PRB (-20°C, red cap) – thaw at room temp, vortex to mix
 - PRE (-20°C, red cap) – store until needed, flick to mix
- ☐ Quick spin
- ☐ Preheat thermocycler (PRB_REM)

1.3 Remove Probes

- ☐ In 1.5 mL tube on ice, prepare Probe Removal Master Mix (see excel template)
 - PRB: 7.7 μ L / sample
 - PRE: 3.3 μ L / sample
 - Pipette MM to mix
- ☐ Add 10 μ L Probe Removal MM to each well
- ☐ Pipette to mix (30 μ L total), seal, quick spin
- ☐ Thermocycler PRB_REM program (32 min)
 - Preheat lid to 100°C
 - 37°C for 15 min
 - 70°C for 15 min
 - Hold at 4°C
 - RNAClean XP (4°C) – bring to room temp for 30 min
 - ELB (rRNA dep. kit, -20°C, clear cap) – thaw at room temp, vortex to mix
 - Prepare 80% EtOH (350 μ L / sample)
- ☐ Quick spin

1.4 Clean Up RNA

- ☐ Vortex RNAClean XP to resuspend
- ☐ Add 60 μ L RNAClean XP beads to each well and mix well by pipette (90 μ L total)
- ☐ Incubate at room temp for 5 min
- ☐ Place samples on magnet for 5 min or until solution appears clear
 - From the cDNA synthesis kit:
 - EPH3 (-20°C, clear cap) – thaw at room temp, vortex to mix
 - FSA (-20°C, brown cap) – thaw at room temp, vortex to mix
 - RVT (-20°C, clear cap) – store until needed, flick to mix

- ☐ Going one column at a time:
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Going one column at a time (if 30 sec has passed since EtOH was added):
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Air dry: Start the timer for 2 min
 - Remove and discard the supernatant from all wells
 - Remove residual liquid with a 10 μ L pipette
 - Allow the pellet to dry, being careful not to over-dry (if the pellet is cracking, then it is too dry). Pellet should appear opaque and slightly shiny.
 - Remove the plate from the magnetic stand
 - Add 10.5 μ L of ELB directly on the beads
- ☐ Resuspend the pellet thoroughly
- ☐ Quick spin to remove content in the lid
- ☐ Incubate for 2 minutes at room temperature
- ☐ Place on magnet for 2 minutes
- ☐ Transfer 8.5 μ L of supernatant to new wells
 - Small amounts of bead carryover do not affect performance
- ☐ Quick spin

2. Fragment and Denature RNA

- ☐ Preheat thermocycler (DEN_RNA)
- ☐ Add 8.5 μ L EPH3 to each well
- ☐ Pipette to mix (17 μ L total), seal, quick spin
- ☐ Thermocycler DEN_RNA program (5min)
 - Preheat lid to 100°C
 - 94°C for 2 min
 - Hold at 4°C
- ☐ Quick spin

3. Synthesize First Strand cDNA

- ☐ Preheat thermocycler (FSS)

- ☐ In 1.5mL tube on ice, prepare First Strand Synthesis Master Mix (see excel template)
 - FSA: 9 μ L / sample
 - RVT: 1 μ L / sample
 - Pipette MM to mix
- ☐ Add 8 μ L First Strand Synthesis MM to each well
- ☐ Pipette to mix (25 μ L total), seal, quick spin
- ☐ Thermocycler FSS program (45min)
 - Preheat lid to 100°C
 - 25°C for 10 min
 - 42°C for 15 min
 - 70°C for 15 min
 - Hold at 4°C
 - From the cDNA synthesis kit:
 - SMM (-20°C, clear cap) – thaw on ice, invert to mix
- ☐ Quick spin
- ☐ Preheat immediately thermocycler with lid open (SSS)

4. Synthesize Second Strand cDNA

4.1 Generate cDNA

- ☐ Add 25 μ L SMM to each well
- ☐ Pipette to mix (50 μ L total), seal, quick spin
- ☐ Thermocycler SSS program (1hr)
 - Preheat lid to 40°C
 - 16°C for 1 hour
 - Hold at 4°C for $\leq$ 16 hours
 - AMPure XP (4°C) – bring to room temp for 30 min
 - RSB (cDNA synthesis kit, -20°C) – thaw at room temp, vortex and invert to mix
 - Prepare 80% EtOH (350 μ L / sample)
- ☐ Quick spin

4.2 Clean Up cDNA

- ☐ Vortex AMPure XP to resuspend
- ☐ Add 90 μL AMPure XP beads to each well and mix well by pipette (140 μL total)
- ☐ Incubate at room temp for 5 min
- ☐ Place samples on magnet for 5 min or until solution appears clear
- ☐ Going one column at a time:
 - ☐ Remove and discard the supernatant
 - ☐ Immediately add 175 μL of fresh 80% EtOH
 - ☐ Proceed to next column and repeat
- ☐ Going one column at a time (if 30 sec has passed since EtOH was added):
 - ☐ Remove and discard the supernatant
 - ☐ Immediately add 175 μL of fresh 80% EtOH
 - ☐ Proceed to next column and repeat
- ☐ Air dry: Start the timer for 2 min
 - ☐ Remove and discard the supernatant from all wells
 - ☐ Remove residual liquid with a 10 μL pipette
 - ☐ Allow the pellet to dry, being careful not to over-dry (if the pellet is cracking, then it is too dry). Pellet should appear opaque and slightly shiny.
 - ☐ Remove the plate from the magnetic stand
 - ☐ Add 19.5 μL of RSB directly on the beads
- ☐ Resuspend the pellet thoroughly
- ☐ Quick spin to remove content in the lid
- ☐ Incubate for 2 minutes at room temperature
- ☐ Place on magnet for 2 minutes
- ☐ Transfer 17.5 μL of supernatant to new wells
 - ☐ Small amounts of bead carryover do not affect performance
- ☐ Quick spin

SAFE STOPPING POINT: -20°C storage for up to 7 days.

5. Adenylate 3' Ends

- ☐ Preheat thermocycler (ATAIL)
- ☐ From the RNA prep ligation kit, thaw:
 - ☐ ATL4 (-20°C, clear cap) – thaw at room temp, flick to mix
- ☐ Add 12.5 μL ATL4 to each well
- ☐ Pipette to mix (30 μL total), seal, quick spin
- ☐ Thermocycler ATAIL program (38min)

- Preheat lid to 100°C
- 37°C for 30 min
- 70°C for 5 min
- Hold at 4°C
 - RSB (cDNA synthesis kit, -20°C) – thaw at room temp, vortex to mix
 - Halfway through ATAIL program, take out AMPure XP beads + anchor plate:
 - Anchor Plate, green (-20°C) – thaw @RT (~10 min.), spin down, shake 1 min. @ 1800 rpm to mix, spin down.

☐ Quick spin

6. Ligate Anchors

☐ Thaw:

- LIGX (RNA prep. kit, -20°C, clear cap) – store until needed, flick to mix
- STL (RNA prep. kit, -20°C, red cap) – thaw at room temp, vortex to mix

☐ Preheat thermocycler (LIG program)

☐ Add in the order specified to each well: (Do not make mastermix)

- 2.5 µL RSB
- 2.5 µL RNA Index Anchors (Seal used wells with foil, indicate “Used”)
- 2.5 µL LIGX

*****NOTE :** if sample input > 100 ng, do not add RSB and add 5 µL RNA Index Anchors instead.

☐ Quick spin

☐ Pipette to mix (38 µL total), seal, quick spin

☐ Thermocycler LIG program (13min)

- Preheat lid to 100°C
- 30°C for 10 min
- Hold at 4°C

☐ Quick spin

☐ Add 5 µL STL to each well

☐ Pipette 15 times to mix (43 µL total), seal, quick spin

7. Clean Up Fragments

- ☐ Prepare 80% EtOH (350 μ L / sample)
- ☐ Vortex AMPure XP to resuspend
- ☐ Add 34 μ L AMPure XP beads to each well and mix well by pipette (77 μ L total)
- ☐ Incubate at room temp for 5 min
- ☐ Place samples on magnet for 5 min or until solution appears clear
 - If proceeding to PCR after the clean up:
 - Index adaptor plate (-20°C) – thaw @ RT, spin down, shake for 1 min. @ 1800 rpm to mix, spin down.
 - EPM (RNA prep. kit, -20°C, clear cap) – thaw at room temp, invert to mix
- ☐ Going one column at a time:
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Going one column at a time (if 30 sec has passed since EtOH was added):
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Air dry: Start the timer for 2 min
 - Remove and discard the supernatant from all wells
 - Remove residual liquid with a 10 μ L pipette
 - Allow the pellet to dry, being careful not to over-dry (if the pellet is cracking, then it is too dry). Pellet should appear opaque and slightly shiny.
 - Remove the plate from the magnetic stand
 - Add 22 μ L of RSB directly on the beads
- ☐ Resuspend the pellet thoroughly
- ☐ Quick spin to remove content in the lid or on sides. Mix by pipette afterwards if spun.
- ☐ Incubate for 2 minutes at room temperature
- ☐ Place on magnet for 2 minutes
- ☐ Transfer 20 μ L of supernatant to new wells
- ☐ Quick spin

SAFE STOPPING POINT: -20°C storage for up to 7 days.

POST-PCR ROOM

8. Amplify Library

- ☐ Preheat thermocycler for PCR, verifying appropriate number of cycles is selected.

PCR cycling recommendations		
Input Amount (ng)	# cycles for RNA	# cycles for FFPE
1	15	17
10	15	17
100	13	15
1000	10	12

- ☐ Thaw:
 - Index adaptor plate (-20°C)- thaw @ RT, spin down, shake for 1 min @ 1800 RPM to mix, spin down.
 - EPM (RNA prep. kit, -20°C, clear cap) – thaw at room temp, invert to mix
- ☐ Pierce the foil covering the index adapter plate wells that will be used (P200)
- ☐ Add in the order specified to each well
 - 10 µL index adapter (UDP0XXX), (Seal used wells with foil, indicate “Used”)
 - Note which index was used for each sample
 - 20 µL EPM
- ☐ Seal plate, spin down, pipette to mix (50 µL total), seal plate, spin down.
- ☐ Thermocycler PCR program (39-45min)
 - Preheat lid to 100°C
 - 98°C for 30 sec
 - 13-15 cycles (15-17 for FFPE); see table
 - 98°C for 10 sec
 - 60°C for 30 sec
 - 72°C for 30 sec

- 72°C for 5 min
- Hold at 4°C for ≤ 16 hours
- ☐ Quick spin
 - Take out AMPure XP beads (4°C) with 30 min. left on PCR. Vortex and invert to mix.
 - Thaw RSB (-20°C), vortex and invert to mix.
 - Prepare 80% EtOH (350 μ L / sample) near the end of the PCR.

SAFE STOPPING POINT: -20°C storage for up to 7 days.

9. Clean Up Library

- ☐ Bring D1000 tape and reagents, and Qubit dsDNA BR kit to room temp 30 min
- ☐ Vortex AMPure XP to resuspend
- ☐ Add 50 μ L AMPure XP beads to each well and mix well by pipette (100 μ L total)
- ☐ Incubate at room temp for 5 min
- ☐ Place samples on magnet for 5 min or until solution appears clear
- ☐ Going one column at a time:
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Going one column at a time (if 30 sec has passed since EtOH was added):
 - Remove and discard the supernatant
 - Immediately add 175 μ L of fresh 80% EtOH
 - Proceed to next column and repeat
- ☐ Air dry: Start the timer for 2 min
 - Remove and discard the supernatant from all wells
 - Remove residual liquid with a 10 μ L pipette

- Allow the pellet to dry, being careful not to over-dry (if the pellet is cracking, then it is too dry). Pellet should appear opaque and slightly shiny.
- Remove the plate from the magnetic stand
- Add 17 μ L of RSB directly on the beads
- ☐ Resuspend the pellet thoroughly
- ☐ Quick spin to remove content in the lid
- ☐ Incubate for 2 minutes at room temperature
- ☐ Place on magnet for 2 minutes
- ☐ Transfer 15 μ L of supernatant to new wells
- ☐ Quick spin

SAFE STOPPING POINT: -20°C storage for up to 30 days.

10. Assess Library Quality (QC)

- ☐ Bring D1000 tape and reagents, and Qubit dsDNA BR kit to room temp 30 min
- ☐ Mix Libraries by shaking at 1800 rpm for 1 min then quickspin, before doing QC

Tapestation

- ☐ Take out Tapestation strip (≤ 15 samples) or 96-well plate (16-95 samples)

- The first well selected is assigned to ladder (can be changed later in software)
- Vortex Sample Buffer and Ladder, then quickspin
- Add 3µl of Sample Buffer to each well containing Ladder or samples
- Add 1µl Sample or Ladder to designated wells. Mix by pipetting up and down
- Quickspin then shake at 2000rpm for 1min. Quickspin one more time
- Put the appropriate adapter (Strip/Plate)
- Insert D1000 tape
 - Once open, tape is good for 14 days. NEVER use open tape older than 14 days
 - On foil, indicate date and number of lanes left.
- Select wells for experiment
 - Wells can only be selected in pairs
- Load samples in the TapeStation then remove lids/seal. Ensure sample is still at bottom
- Ensure tips are loaded and tip waste is empty
- Start run
- Check that the electropherogram shows an average fragment length of 375-475 bp.

Qubit

- Label Qubit tubes (# samples + 2 standards)
- Vortex Dye, Standard 1 and Standard 2, then quickspin
- Prepare MasterMix (# samples + 2 standards + extra) by diluting Dye 1/200 using Buffer
 - Follow table taped to MasterMix bench
- Add MasterMix to Qubit tubes:
 - 190µl for Standard (MasterMix bench)
 - 198µl for Samples (MasterMix bench)
- Add Standard/Sample to Qubit tubes:
 - 10µl for Standard (Lib Prep bench)
 - 2µl for Samples (Lib Prep bench)
- Vortex Qubit tubes
- Incubate for 2 min at room temp
- Read